

Python Tasks – Day16

1. Write a program to implement these formulae of permutations and combinations.

Number of permutations of n objects taken r at a time is: function name permutation

$$p(n,r) = n! / (n-r)!$$

Number of combinations of n objects taken r at a time is: function name combination

$$c(n,r) = n! / (r! * (n-r)!))$$

$$\text{or } c(n,r) = p(n,r) / r!$$

2. Write a python function `prime_factor` to find out the prime factors of a number. Some examples of prime factors are –

Prime factors of 56 – 2, 2, 2, 7

Prime factors of 98 – 2, 7, 7

Prime factors of 121 – 11, 11

3. Write a function `cubesum()` that accepts an integer and returns the sum of the cubes of individual digits of that number
4. Write a function `is Leap()` which inputs a year and return 1 if the year is leap otherwise 0.
5. Two different numbers are called amicable number if the sum of the proper divisors of each is equal to the other number. For example 220 and 284 are amicable numbers.

Sum of proper divisors of 220=1+2+4+5+10+11+20+22+44+55+110=284

Sum of proper divisors of 284=1+2+4+71+142=220.

Write a function to print pairs of amicable numbers in a range.